PROBLEM 11 | Problems T2_CrystStruct.pdf | Crystalline Structures & Miller Indices

Lattice Parameter and Densities of Aluminium from Linear Density

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11. Aluminum has an FCC structure with atomic weight of $26.98\ \mathrm{g/mol}$***. Given that its linear density in the direction*** $[111]$ ***is*** $1.43 \times 10^{7}\ \mathrm{at/cm}$***, determine: a)*** *The lattice parameter and the atomic radius for aluminum. (Solution:* $a = 4.04\ \text{Å}$*,* $r = 1.43\ \text{Å}$*) **b)** Volume density in* $(\mathrm{g/cm^3})$ *and planar density in the plane* (101) *in* $(\mathrm{at/cm^2})$*. (Solution:* $\rho = 2.71\ \mathrm{g/cm^3}$*;* $\rho_{(101)} = 8.66 \times 10^{14}\ \mathrm{at/cm^2}$*)*

Solution: $a = 4.04\ \text{Å},\ r = 1.43\ \text{Å}$

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Element: Pure aluminium (Al).
- Structure: Face-Centred Cubic (FCC) $\implies n = 4$ atoms/cell.
- Molar atomic mass: $M = 26.98\ \mathrm{g/mol}$.
- Linear density in the $[111]$ direction: $\rho_{[111]} = 1.43 \times 10^{7}\ \mathrm{at/cm}$.
- Avogadro number: $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Linear Density on $[111]$ for FCC [Session 4 Slide 8]:**

In an FCC cell, the main diagonal $[111]$ has length $L_{[111]} = a\sqrt{3}$. It passes through only two atoms located at the opposite end vertices, each contributing $\frac{1}{2}$ centred atom:

$$N_{\text{atoms}} = 2 \times \frac{1}{2} = 1\ \text{atom}$$

$$\rho_{[111]} = \frac{1}{a\sqrt{3}} \implies a = \frac{1}{\sqrt{3} \cdot \rho_{[111]}}$$

2. **Atomic Radius in FCC [Session 3 Slide 27]:**

$$4R = a\sqrt{2} \implies R = \frac{a\sqrt{2}}{4}$$

3. **Volumetric Density (ρ_v):**

$$\rho_v = \frac{n \cdot M}{a^3 \cdot N_A} = \frac{4 \cdot M}{a^3 \cdot N_A}$$

4. **Planar Density on the (101) Plane [Session 4 Slide 9]:**

In the cubic system, the (101) plane is crystallographically equivalent to $(110) \in \{110\}$.

The (101) plane cuts the diagonal of the $x - z$ face (length $a\sqrt{2}$) and the y edge (length a).

Area: $A_{(101)} = a\sqrt{2} \times a = a^2\sqrt{2}$.

Atoms contained: $N_{\text{atoms}} = 4 \times \frac{1}{4} + 2 \times \frac{1}{2} = 2$ atoms.

$$\rho_{(101)} = \frac{2}{\sqrt{2}a^2} = \frac{\sqrt{2}}{a^2}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Lattice Parameter and Atomic Radius

- **Calculation of a:**

$$a = \frac{1}{\sqrt{3} \times (1.43 \times 10^7 \text{ at/cm})} = \frac{1}{1.73205 \times 1.43 \times 10^7 \text{ cm}^{-1}} = \frac{1}{2.4768 \times 10^7 \text{ cm}^{-1}}$$

$$a = 4.0374 \times 10^{-8} \text{ cm} \approx \mathbf{4.04} \text{ Å}$$

- **Calculation of R:**

$$R = \frac{a\sqrt{2}}{4} = \frac{(4.0374 \times 10^{-8}\ \text{cm}) \times 1.4142136}{4} = 1.4274 \times 10^{-8}\ \text{cm} \approx \mathbf{1.43}\ \text{Å}$$

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2. Part b: Volumetric and Planar Density

- **Volumetric Density (ρ_v):**

$$V_C = a^3 = (4.0374 \times 10^{-8}\ \text{cm})^3 = 6.5813 \times 10^{-23}\ \text{cm}^3$$

$$\rho_v = \frac{4 \times 26.98\ \text{g/mol}}{(6.5813 \times 10^{-23}\ \text{cm}^3) \times (6.022 \times 10^{23}\ \text{mol}^{-1})} = \frac{107.92}{39.633} = \mathbf{2.72}\ \text{g/cm}^{\mathbf{3}} \approx \mathbf{2.71}\ \text{g/cm}^{\mathbf{3}}$$

- **Planar Density on the (101) Plane:**

$$a^2 = (4.0374 \times 10^{-8}\ \text{cm})^2 = 1.63006 \times 10^{-15}\ \text{cm}^2$$

$$\rho_{(101)} = \frac{\sqrt{2}}{a^2} = \frac{1.4142136}{1.63006 \times 10^{-15}\ \text{cm}^2} = 8.6758 \times 10^{14} \approx \mathbf{8.66 \times 10^{14}}\ \text{at/cm}^{\mathbf{2}}$$

`PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION`

- **Cross-Validation of Methods:** In problem T2-CS05 the planar density was obtained from the known atomic radius $R = 1.43$ Å. Here, the procedure is reversed: by measuring the linear density $\rho_{[111]}$ (accessible through high-resolution transmission electron microscopy or diffraction), the lattice parameter (4.04 Å), the atomic radius (1.43 Å), the density of the solid ($2.71\ \text{g/cm}^3$) and the planar density of the shear planes are derived with identical accuracy.